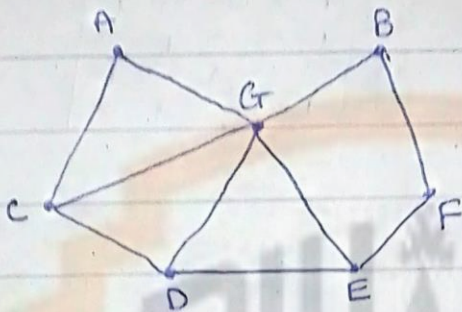


Graphs

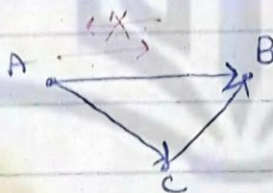
Definition: "A graph is a collection of vertices and edges."

e.g.

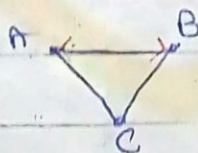


There are two types of edges.

i) Directed

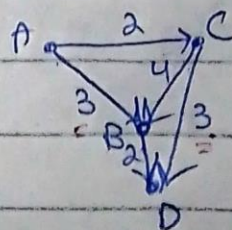
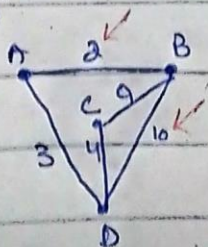


ii) UnDirected



These edges bear a weight, by default it's considered "1", else it must be mentioned.

e.g.

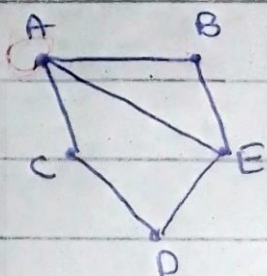


Two edges can have same weight.

Graph Representation

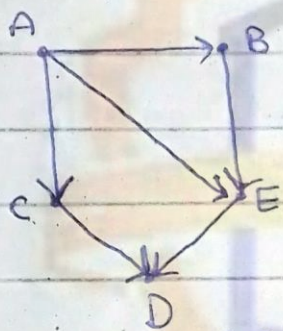
1 Matrix

a. Adjacency Matrix.



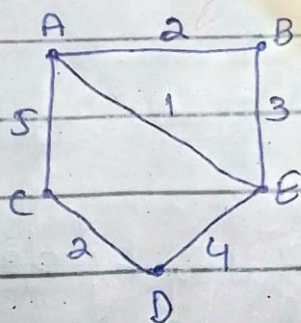
Simple Graph.

	A	B	C	D	E
A	0	1	1	0	1
B	1	0	0	0	1
C	1	0	0	1	0
D	0	0	1	0	1
E	1	0	0	1	0



Directed Graph.

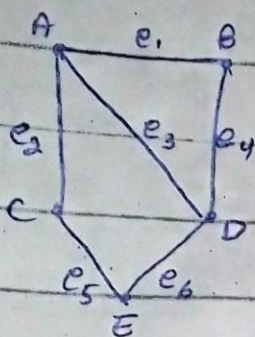
	A	B	C	D	E
A	0	1	1	0	1
B	0	0	0	0	1
C	0	0	0	1	0
D	0	0	0	0	0
E	0	0	0	1	0



	A	B	C	D	E
A	0	2	5	0	1
B	2	0	0	0	3
C	5	0	0	2	0
D	0	0	2	0	4
E	1	3	0	4	0

Weighted.

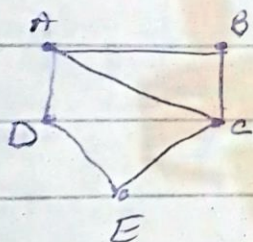
b. Incidence Matrix.



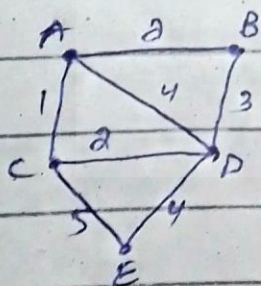
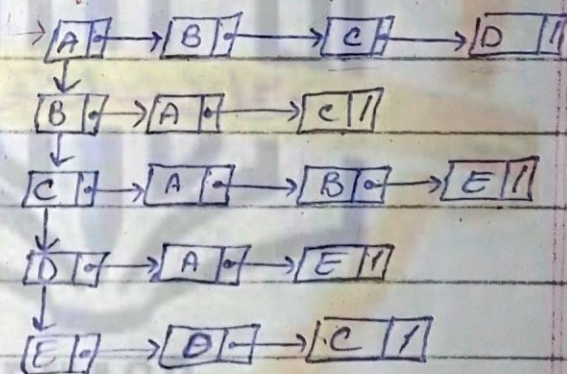
	A	B	C	D	E
e_1	1	1	0	0	0
e_2	1	0	1	0	0
e_3	1	0	0	1	0
e_4	0	1	0	1	0
e_5	0	0	1	0	1
e_6	0	0	0	1	1

2. List

Adjacency list.



simple Graph



weighted

